

Akshat Raj

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SUMMARY

Data Analyst with hands-on internship experience in enterprise IT and e-commerce analytics. Proficient in Python, SQL, Power BI, and Excel for building ETL pipelines, KPI dashboards, and RFM-based customer segmentation. Additional background in LLM engineering and multi-agent AI systems. Pursuing B.Tech in Computer Science (AI/ML) at VIT Bhopal; graduating May 2027.

TECHNICAL SKILLS

Languages & Query	Python, JavaScript (Node.js), SQL (MySQL, PostgreSQL, T-SQL)
AI & LLM Engineering	LangChain.js, LangGraph, GPT-4o API, Prompt Engineering, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), OpenAI Embeddings, Structured Outputs
Databases	MongoDB, ChromaDB (Vector Database), MySQL, PostgreSQL, Microsoft SQL Server (SSMS)
Backend & APIs	Express.js, REST APIs, Mongoose, Tavily Search API, Slack API, node-cron
Frontend	React.js, Next.js, Tailwind CSS
BI & Analytics	Power BI (DAX, Power Query), Microsoft Excel, Tableau, Data Visualization, KPI Development
Developer Tools	Git, GitHub, Docker, Docker Compose, Railway, VS Code, Postman, Jupyter Notebook
Core Competencies	ETL Pipelines, Data Warehousing, Dimensional Modeling (Star Schema, Medalion Architecture), Semantic Search, Workflow Automation, Dashboard Design, Data Storytelling, Query Optimization, Agile Methodology

EXPERIENCE

Data Analyst Intern <i>CESC Rajasthan, RP-Sanjiv Goenka Group</i>	May 2026 – Present <i>Rajasthan, India</i>
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- Spearheaded data analysis and reporting across 10+ enterprise IT workflows, transforming raw operational datasets into executive-level Power BI dashboards for senior management, reducing manual consolidation time by **35%** per reporting cycle.
- Built Python (Pandas) and Excel ETL pipelines to automate KPI tracking and report generation, eliminating **8+ hours** of manual effort per cycle and improving cross-departmental data accuracy.
- Coordinated with IT, Finance, and Operations teams to translate business requirements into structured analytical deliverables under tight enterprise deadlines, improving stakeholder turnaround by **20%**.
- Conducted root cause analysis and exploratory data analysis (EDA) on operational datasets, surfacing **3 critical process inefficiencies** that directly informed IT workflow improvements.

Data Analyst Intern <i>Thrillophilia</i>	Apr 2025 – Jun 2025 <i>Remote, India</i>
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- Processed and analyzed **100K+ booking records** using Python (Pandas, NumPy) and SQL; performed data validation and exploratory analysis to identify trends in travel package performance.
- Developed a multi-page **Power BI dashboard** tracking revenue trends, regional performance, and booking funnels, consolidating reports previously spread across multiple manual Excel files.
- Executed **RFM segmentation** on customer data to classify cohorts by recency, frequency, and monetary value, delivering a prioritized re-engagement target list to the marketing team.
- Designed and analyzed an **A/B test** on two landing page variants using hypothesis testing (p-value, confidence intervals), providing statistical evidence to guide the conversion optimization decision.
- Automated weekly KPI summary reports via Python and Excel Power Query, eliminating repetitive manual work from the recurring reporting workflow.

PROJECTS

Retail Analytics Pipeline — <i>Python, Pandas, Matplotlib, openpyxl</i> / <i>GitHub</i>	Feb 2026
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- Engineered a scalable pipeline processing **32.4M+ records** from 3 heterogeneous sources; reduced memory footprint by **65%** through dtype-level optimization with full Git version control.
- Built a repeatable data cleaning framework resolving nulls, duplicates, and encoding errors; produced a structured data

- quality report establishing audit trails for governance compliance.
- Computed **13 business KPIs** including INR 818 Cr revenue and 32.44% gross margin; identified 2 departments driving **44% of total revenue**, enabling data-backed category prioritization.
- Scored **206,209 customers** on RFM dimensions; Champions cohort showed **10x higher spend** vs. Lost segment, with 24,123 at-risk customers flagged as the highest-ROI re-engagement target.

SQL Data Warehouse & Analytics Pipeline — *SQL Server, T-SQL, SSMS, Git | GitHub* Jul 2026

- Architected a 3-layer **Medallion (Bronze–Silver–Gold) data warehouse** in SQL Server, transforming raw CRM and ERP data into analytics-ready datasets via structured ETL pipelines.
- Designed **dimensional data models** (Fact and Dimension tables, Star Schema) and built **stored procedures, views, CTEs, and window functions** in T-SQL for automated cleansing, deduplication, and validation.
- Authored analytical SQL queries to generate business KPIs, applying indexing and query-optimization techniques; version-controlled the full pipeline via Git and GitHub.

AI Competitive Intelligence Agent — *LangChain.js, LangGraph, GPT-4o, Node.js, Express.js, React.js, Next.js, MongoDB, ChromaDB, Docker* Jun 2026

- Designed an autonomous **multi-agent AI system** that continuously monitors competitor websites, pricing pages, blogs, and job boards using a **4-agent LangGraph workflow** with structured outputs.
- Engineered a self-updating **RAG pipeline** using ChromaDB and OpenAI Embeddings with semantic retrieval and deduplication to maintain a fresh, noise-free knowledge base.
- Integrated Tavily Search, GPT-4o, and Slack APIs to deliver automated strategic briefings and real-time competitive alerts to stakeholders.
- Built a React/Next.js + Express.js dashboard backed by MongoDB for competitor management and briefing history; implemented an evaluation framework measuring retrieval precision and alert accuracy.

EDUCATION

VIT Bhopal University Aug 2023 – May 2027
B.Tech, Computer Science and Engineering – Artificial Intelligence and Machine Learning | CGPA: 8.63/10 Bhopal, MP

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Statistical Methods, Machine Learning, Data Mining, Python Programming, Cloud Computing, Operating Systems

G.S. Janghid School, CBSE 2022 – 2023
Senior Secondary, Class XII: 83% Jodhpur, Rajasthan

St. Anne's School, CBSE 2020 – 2021
Secondary, Class X: 90% Jodhpur, Rajasthan

CERTIFICATIONS

Google Data Analytics Professional Certificate, Coursera (Google) – 2026
 Data Analysis, SQL, Tableau, R

Data Science Bootcamp, Krish Naik – Python, ML, EDA, Feature Engineering, Model Deployment 2025

Applied Machine Learning in Python, Coursera, University of Michigan 2025
 – Scikit-learn, Model Evaluation

Cloud Computing, NPTEL, IIT Kharagpur – Cloud Architecture, GCP, Distributed Systems 2025